

Emile Pierret

Born on January 21, 1997.

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🌐 <https://emilepi.github.io>

Education

- 2025 – Present 📖 **Post-doc.** *Université Paris Cité, ENS, France*
Advisor: Julie Delon.
- 2022 – 2025 📖 **Ph.D. in Mathematics.** *University of Orléans, France*
Thesis topic: *Stochastic Super-resolution and Inverse Problems: From Gaussian Conditional Sampling to Diffusion Models.*
Ph.D. advisor: Bruno Galerne.
- 2021 – 2022 📖 **Master's Degree (M2) in Mathematics, Vision, Learning (MVA).** *ENS Paris-Saclay, Gif-sur-Yvette, France*
Research-oriented Master's program.
- 2020 – 2021 📖 **French National Agrégation in Mathematics (external track).** *ENS Paris-Saclay, Gif-sur-Yvette, France*
Option B.
- 2019 – 2020 📖 **Master's Degree (M1) – Jacques Hadamard Program.** *ENS Paris-Saclay, Cachan, France*
Research internship: *Uncertainty quantification in COVID-19 spread: lockdown effects.*
Supervisor: Ana Carpio, *Universidad Complutense de Madrid.*
- 2018 – 2019 📖 **Double Bachelor's Degree in Mathematics and Computer Science.** *ENS Paris-Saclay, Cachan, France*
- 2018 📖 **Admitted as a normalien at ENS Paris-Saclay (4-year program)**
Admission via the Computer Science competitive entrance exam.
- 2015 – 2018 📖 **Preparatory Classes for the French Grandes Écoles.** *Lycée Descartes, Tours, France*
MPSI (Mathematics, Physics, Engineering Science), followed by MP* (Advanced Mathematics and Physics).

Publications

Preprints

- 1 E. Pierret, V. Tosel, J. Delon, and A. Newson, *Flow matching for applied mathematicians*. 2026.
- 2 É. Pierret and B. Galerne, *Exact evaluation of the accuracy of diffusion models for inverse problems with gaussian data distributions*, 2025.

Journal papers

- 1 É. Pierret and B. Galerne, “Stochastic super-resolution for gaussian microtextures,” *SIAM Journal on Imaging Sciences*, vol. 18, no. 2, pp. 1176–1207, 2025. 📄 DOI: 10.1137/24M1657407.
- 2 A. Carpio and E. Pierret, “Uncertainty quantification in covid-19 spread: Lockdown effects,” *Results in Physics*, vol. 35, p. 105 375, 2022, ISSN: 2211-3797. 📄 DOI: <https://doi.org/10.1016/j.rinp.2022.105375>.

Conference papers

- 1 E. Pierret, J. Hertrich, S. Hurault, and J. Delon, “Tessellations of semi-discrete flow matching,” in *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026.
- 2 E. Pierret and B. Galerne, “Diffusion models for gaussian distributions: Exact solutions and wasserstein errors,” in *Forty-second International Conference on Machine Learning*, 2025.  URL: <https://openreview.net/forum?id=bxYbxzCI2R>.
- 3 É. Pierret and B. Galerne, “Stochastic super-resolution for gaussian textures,” in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.  DOI: 10.1109/ICASSP49357.2023.10096585.

Talks

Invited Talks

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| December 2026 |  From Diffusion Models to Flow Matching: A Gentle Introduction to Modern Generative Sampling.
<i>Journée du projet CaSciModOT</i> , University of Orléans. |
| June 2025 |  A Precise Examination of Diffusion Models via Their Application to Gaussian Distributions.
<i>Images, Optimization and Probability Seminar</i> , University of Bordeaux. |
| June 2025 |  On the accuracy of diffusion models in Bayesian image inverse problems: A Gaussian case study.
<i>SMAI 2025</i> , Carcans Maubuisson. |
| May 2025 |  On the accuracy of diffusion models in Bayesian image inverse problems: A Gaussian case study.
<i>ANR MISTIC days</i> , Lyon. |
| February 2025 |  Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.
<i>Mathematical Imaging and Surface Processing Workshop</i> , MFO Oberwolfach, Germany. |
| January 2025 |  Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.
<i>Imaging in Paris Seminar</i> , Paris. |
| November 2024 |  Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.
<i>Workshop on Stochastic Geometry and Mathematics for Imaging</i> , Nice. |
| October 2024 |  Stochastic super-resolution for Gaussian microtextures.
<i>ANR MISTIC days</i> , Vannes. |
| June 2024 |  Stochastic super-resolution for Gaussian microtextures.
<i>LAREMA PhD Seminar</i> , Angers. |

Talks (continued)

- May 2024  **Introduction to diffusion models and their restriction to the Gaussian case.**
CANUM 2024, Île de Ré.

Posters

- July 2025  **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.**
Poster presentation, *ICML 2025*, Vancouver.
- January 2025  **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.**
Poster presentation, *Mathematics and Image Analysis (MIA'25)*, Paris.
- June 2023  **Stochastic super-resolution for Gaussian textures.**
Poster presentation, *ICASSP 2023*, Rhodes, Greece.

Outreach Talks and Working Group Presentations

- October 2026  **A Precise Examination of Diffusion Models via Their Application to Gaussian Distributions .**
CSD seminar, ENS Paris.
- June 2024  **Tutorial: *Introduction to Neural Networks*.**
IDP PhD Week, Courcimont Farm.
- June 2023  **Introduction to diffusion models and practical session.**
IDP Deep Learning Working Group, Orléans.
- June 2023  **Presentation of the “RePaint” method.**
Diffusion Models Working Group, MAP5, Paris.
- June 2023  **Stochastic super-resolution for Gaussian textures.**
IDP PhD Week, Courcimont Farm.

Teaching Experience

- 2022 – 2025  **Lectures and practicals – Image Learning (15h).** *University of Orléans*
Master’s level (M1) in Applied Mathematics.
Introductory course on neural networks for image classification.
- 2023 – 2025  **Tutorials – Algebra 3 (49h).** *University of Orléans*
Second-year undergraduate level (L2) in Mathematics.
Tutorials in linear algebra.
- 2022 – 2023  **Tutorials – Calculus (49h).** *University of Orléans*
First-year undergraduate level (L1), first semester.
Tutorials introducing fundamental tools in analysis and algebra.

Service and Responsibilities

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| 2024 | 📌 | Elected alternate member of the Research Committee , University of Orléans |
| 2022–2025 | 📌 | Organizer of the PhD seminar , Institut Denis Poisson (Orléans–Tours) |

Skills

Languages

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| English | 📌 | C1 level (IELTS 7/9, 2020) |
| Spanish | 📌 | C1 level |

Programming

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| Advanced proficiency | 📌 | Python, PyTorch, MATLAB, $\text{\LaTeX}$, Caml |
| Basic knowledge | 📌 | C, C++, Assembly language, R |